Topic 7 — Electric and Magnetic Fields

7.1 Electric Fields

An electric field is a region of space where a charged particle experiences a force. Electric field lines show the direction of force on a positive test charge. Field lines never cross, and their density indicates the field strength.

Radial Fields

A **radial field** is produced by a point charge (or a charged sphere). The field lines radiate outward from a positive charge and inward toward a negative charge. The field strength decreases with distance from the charge according to an inverse-square law, meaning that doubling the distance reduces the field strength to one quarter.

Uniform Fields

A **uniform field** exists between two parallel charged plates. The field lines are parallel and equally spaced, indicating that the field strength is the same at all points between the plates (except near the edges where fringing occurs).

$$E = V / d$$

where V is the potential difference between the plates (V) and d is the separation (m). This equation only applies to a uniform field.

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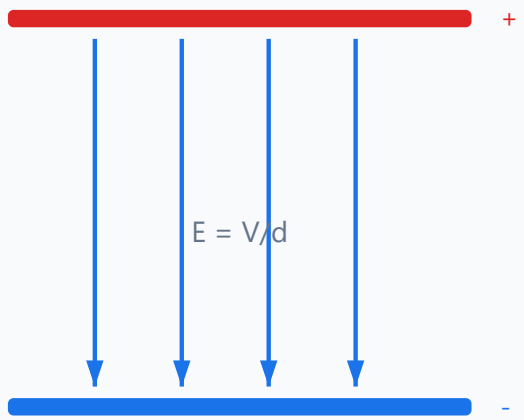
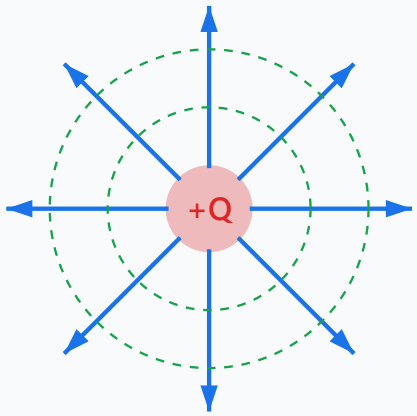
Electric Field Patterns

Uniform Field (Parallel Plates)



Radial Field (Point Charge +Q)



Equipotentials (dashed green)

Coulomb's Law

The force between two point charges is given by Coulomb's law:

$$F = kQ_1Q_2 / r^2 = (1 / 4\pi\varepsilon_0) \times Q_1Q_2 / r^2$$

where $k = 1/(4\pi\varepsilon_0) = 8.99 \times 10^9$ N m^2 C^{-2} and $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} is the permittivity of free space.

The force is attractive between unlike charges and repulsive between like charges. The magnitude of the force follows an inverse-square law with distance.

Worked Example: A proton and an electron are separated by 5.3×10^{-11} m (the Bohr radius of hydrogen). Calculate the electrostatic force between them.

$F = kQ_1Q_2 / r^2 = (8.99 \times 10^9)(1.60 \times 10^{-19})^2 / (5.3 \times 10^{-11})^2$
$F = (8.99 \times 10^9)(2.56 \times 10^{-38}) / (2.81 \times 10^{-22})$
$F =$ **8.2×10^{-8} N**

For comparison, the gravitational force between the same particles is approximately 3.6×10^{-47} N, making the electrostatic force about 10^{39} times stronger.

Electric Field Strength

Electric field strength (E) at a point is the force per unit positive charge placed at that point.
E = F / Q (unit: N C^{-1} or V m^{-1})

For a radial field around a point charge Q:

$$E = kQ / r^2 = Q / (4\pi\varepsilon_0 r^2)$$

For a uniform field between parallel plates:

$$E = V / d$$

Worked Example: Two parallel plates are separated by 2.0 cm with a potential difference of 500 V across them. Calculate the electric field strength and the force on an electron between the plates.

E = V / d = 500 / 0.020 = **25 000 V m^{-1}** = 25 kV m^{-1}

F = EQ = 25 000 × 1.60 × 10^{-19} = **4.0 × 10^{-15} N**

Electric Potential

Electric potential (V) at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point. It is a scalar quantity.

$$V = kQ / r = Q / (4\pi\varepsilon_0 r)$$

Key points about electric potential:

- Potential is a **scalar** quantity (no direction), so potentials from multiple charges add algebraically
- Potential is zero at infinity
- Positive charges create positive potential; negative charges create negative potential

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- Work done moving charge q through potential difference ΔV: $W = q\Delta V$
- The electronvolt (eV) is the energy gained by an electron accelerated through 1 V: 1 eV = 1.60×10^{-19} J

Equipotentials

Equipotential lines (or surfaces) connect points of equal potential. They are always perpendicular to field lines. No work is done moving a charge along an equipotential because there is no component of the electric force in that direction.

> **Exam Tip:** The electric field strength is the negative of the potential gradient: $E = -dV/dr$. On a V-r graph, the gradient at any point gives E at that distance. A steeper gradient means a stronger field.

Comparison of Electric and Gravitational Fields

Property	Electric Field	Gravitational Field
Source	Charges	Masses
Force law	$F = kQ_1Q_2/r^2$	$F = GM_1M_2/r^2$
Nature	Attractive or repulsive	Always attractive
Field strength	$E = F/Q$ (N C^{-1})	$g = F/m$ (N kg^{-1})
Potential	$V = kQ/r$ (can be + or −)	$V = -GM/r$ (always negative)
Relative strength	Very strong	Extremely weak

7.2 Capacitance

Capacitance (C) is the charge stored per unit potential difference across the capacitor. It measures the ability of a component to store charge.

$C = Q / V$ (unit: farad, F; 1 F = 1 C V^{-1})

A 1 farad capacitor is enormously large. Practical capacitors are typically measured in microfarads (μF = 10^{-6} F), nanofarads (nF = 10^{-9} F), or picofarads (pF = 10^{-12} F).

Parallel Plate Capacitor

$$C = \varepsilon_0 \varepsilon_r A / d$$

where A is the area of overlap of the plates (m^2), d is the separation (m), ε_0 is the permittivity of free space, and ε_r is the relative permittivity (dielectric constant) of the material between the plates.

A **dielectric** is an insulating material placed between the plates. It increases the capacitance by a factor of ε_r because it reduces the effective electric field between the plates (polar molecules in the dielectric partially cancel the applied field).

Capacitors in Combination

Arrangement	Formula	Notes
Parallel	$C_{total} = C_1 + C_2 + C_3 + ...$	Same p.d. across each; charges add
Series	$1/C_{total} = 1/C_1 + 1/C_2 + 1/C_3 + ...$	Same charge on each; p.d.s add

Exam Tip: Note that capacitors in parallel add directly (like resistors in series), while capacitors in series add reciprocally (like resistors in parallel). This is the opposite way round from resistors.

Worked Example: A 47 μF and a 22 μF capacitor are connected in series. Find the total capacitance and the charge stored when connected to a 9.0 V supply.

$1/C_{total} = 1/47 + 1/22 = 0.02128 + 0.04545 = 0.06673\ \mu F^{-1}$

$C_{total} = 1/0.06673 =$ **15.0 μF**

$Q = CV = 15.0 \times 10^{-6} \times 9.0 =$ **1.35×10^{-4} C** = 135 μC

Both capacitors store the same charge (135 μC) because they are in series.

Energy Stored in a Capacitor

The energy stored in a charged capacitor is the area under the Q-V graph (a straight line through the origin):

$$E = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2}Q^2/C$$

Worked Example: A 1000 μF capacitor is charged to 12 V. Calculate the energy stored, and the new voltage if it is then connected to an uncharged 1000 μF capacitor.

Energy stored = $\frac{1}{2}CV^2 = \frac{1}{2}(1000 \times 10^{-6})(12)^2$ = **0.072 J** = 72 mJ

When connected to the uncharged capacitor, charge is shared equally (total C = 2000 μF, total Q = CV = 0.012 C):
New V = Q / C_{total} = 0.012 / 0.002 = **6.0 V**

New energy = $\frac{1}{2}(0.002)(6.0)^2$ = 0.036 J = 36 mJ
Half the original energy has been dissipated as heat in the connecting wires.

Charging and Discharging

When a capacitor charges or discharges through a resistor, the charge, voltage, and current change exponentially.

Discharging:

$$Q = Q_0\, e^{-t/RC} \quad V = V_0\, e^{-t/RC} \quad I = I_0\, e^{-t/RC}$$

Charging:

$$Q = Q_0(1 - e^{-t/RC}) \quad V = V_0(1 - e^{-t/RC})$$

The current during charging still decays exponentially: $I = I_0\, e^{-t/RC}$

Time Constant

The time constant ($\tau = RC$) is the time taken for the charge (or voltage or current) to fall to 1/e (≈ 37%) of its initial value during discharge, or to rise to (1 − 1/e) ≈ 63% of its final value during charging.

After 5 time constants (5RC), the capacitor is considered fully charged or fully discharged (>99%).

Worked Example: A 470 μF capacitor discharges through a 10 kΩ resistor from an initial voltage of 9.0 V. Calculate: (a) the time constant, (b) the voltage after 3.0 s, (c) the time for the voltage to fall to 1.0 V.

(a) $\tau = RC = (10 \times 10^3)(470 \times 10^{-6})$ = **4.7 s**

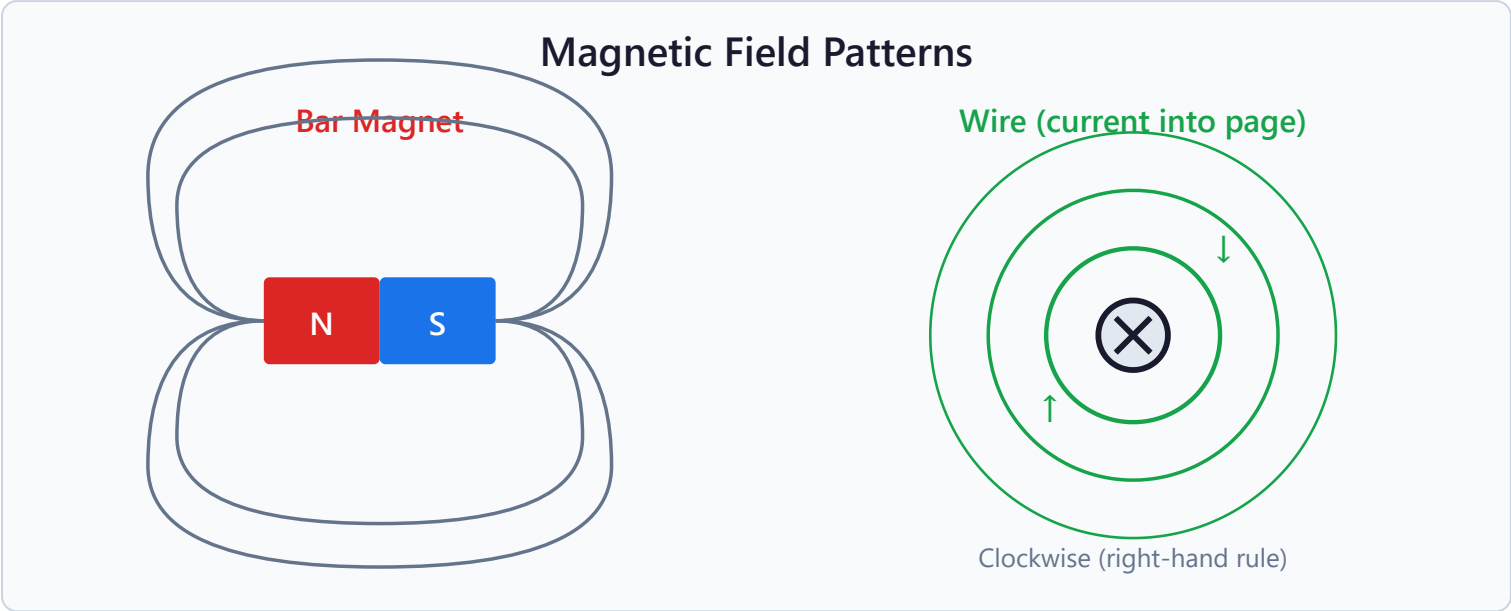
(b) $V = V_0 e^{-t/RC} = 9.0 \times e^{-3.0/4.7} = 9.0 \times e^{-0.638} = 9.0 \times 0.528$ = **4.8 V**

(c) $1.0 = 9.0 e^{-t/4.7}$
$e^{-t/4.7} = 1/9 = 0.111$
$-t/4.7 = \ln(0.111) = -2.197$
$t = 2.197 \times 4.7$ = **10.3 s**

Exam Tip: To find the time constant from a graph, read off the time at which V (or Q) has fallen to 0.37 of its initial value. Alternatively, use ln(V) vs t — the gradient equals $-1/RC$.

7.3 Magnetic Fields

Magnetic flux density (B) is a measure of the strength of a magnetic field. It is defined as the force per unit length per unit current on a conductor carrying current perpendicular to the field.
Unit: tesla (T), where $1\ T = 1\ kg\ s^{-2}\ A^{-1}$



Force on a Current-Carrying Conductor

A straight conductor of length L carrying current I at angle θ to a uniform magnetic field B experiences a force:

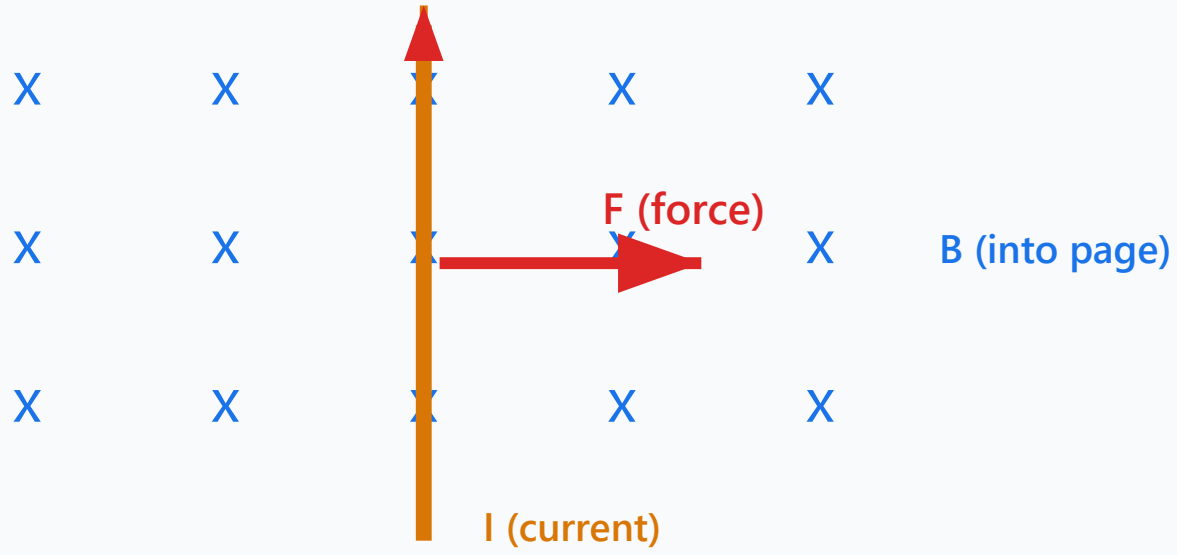
$$F = BIL \sin \theta$$

The direction of the force is given by **Fleming's left-hand rule**:

- **F**irst finger: **F**ield (B)
- **S**econd finger: Current (I, conventional current direction)
- **Thu**mb: **Thu**st (Force on conductor)

When θ = 90° (current perpendicular to field), F = BIL (maximum force). When θ = 0° (current parallel to field), F = 0 (no force).

Fleming's Left-Hand Rule: F = BIL



First finger: Field | Second finger: Current | Thumb: Force

Worked Example: A wire of length 0.25 m carries a current of 3.0 A perpendicular to a magnetic field of flux density 0.15 T. Calculate the force on the wire.

F = BIL sin θ = 0.15 × 3.0 × 0.25 × sin 90° = **0.11 N**

Force on a Moving Charge

A charged particle moving through a magnetic field experiences a force:

$$F = BQv \sin \theta$$

where Q is the charge, v is the velocity, and θ is the angle between v and B.

The force is always perpendicular to the velocity, so it does no work on the charge and cannot change its speed — only its direction. A charged particle moving perpendicular to a uniform magnetic field follows a **circular path**:

$$BQv = mv^2/r \rightarrow r = mv / (BQ)$$

The radius of the circular path increases with momentum and decreases with charge and field strength. This principle is used in cyclotrons, mass spectrometers, and the Large Hadron Collider.

Worked Example: An electron ($m = 9.11 \times 10^{-31}$ kg, $Q = 1.60 \times 10^{-19}$ C) moving at 2.0×10^6 m s^{-1} enters a magnetic field of 0.50 mT perpendicular to its velocity. Find the radius of its circular path.

$r = mv / (BQ) = (9.11 \times 10^{-31} \times 2.0 \times 10^6) / (0.50 \times 10^{-3} \times 1.60 \times 10^{-19})$
$r = (1.822 \times 10^{-24}) / (8.0 \times 10^{-23})$ = **0.023 m** = 2.3 cm

Magnetic Flux

Magnetic flux (Φ) through a surface is the product of the magnetic flux density and the area perpendicular to the field.
$\Phi = BA \cos\theta$ (unit: weber, Wb, where 1 Wb = 1 T m^2)

Flux linkage for a coil of N turns: $N\Phi = BAN \cos\theta$

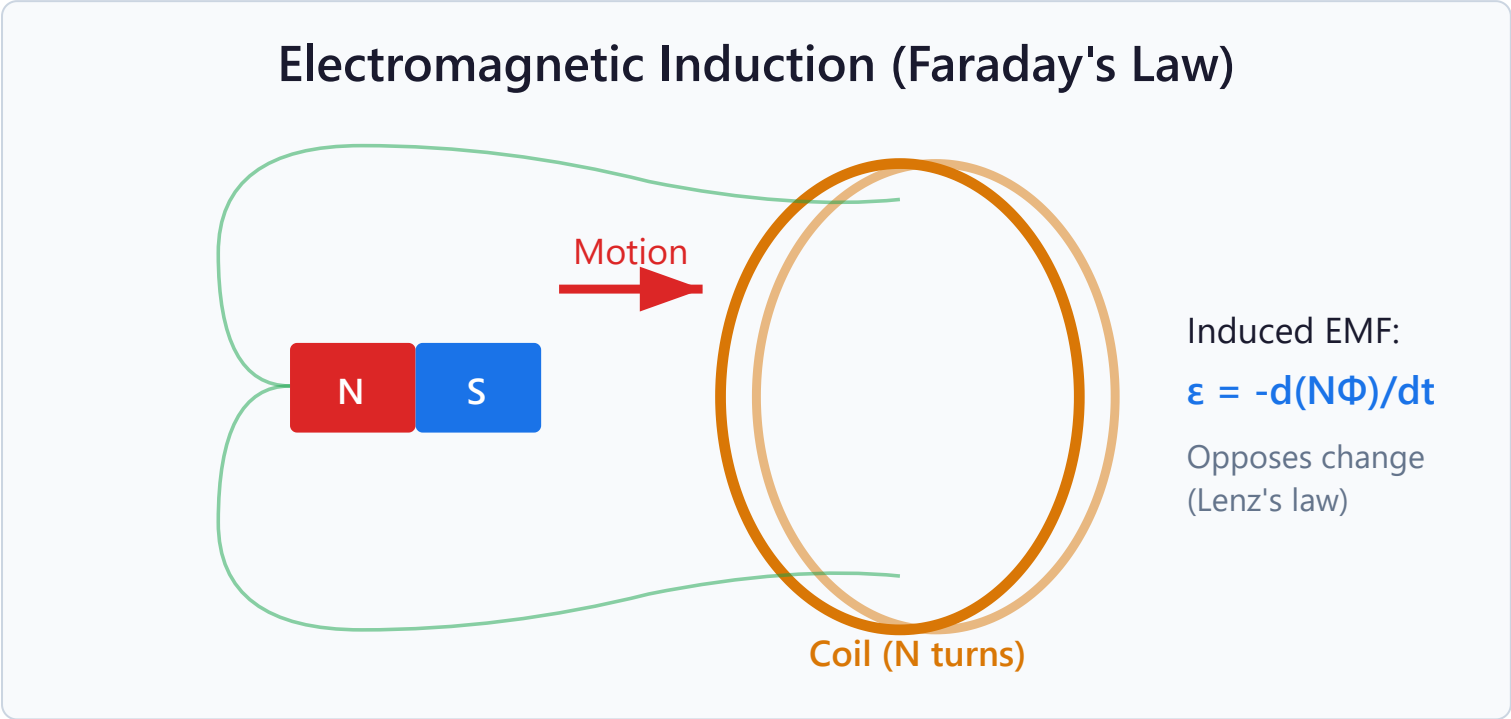
When θ = 0° (field perpendicular to the plane of the coil, i.e., through it), flux linkage is maximum. When θ = 90° (field parallel to the coil plane), flux linkage is zero.

7.4 Electromagnetic Induction

Faraday's law: The magnitude of the induced EMF is equal to the rate of change of magnetic flux linkage through the circuit.
$\varepsilon = -d(N\Phi)/dt$

Lenz's law: The direction of the induced EMF (and hence the induced current) is such that it opposes the change in flux that produces it. This is represented by the negative sign in Faraday's law.

Lenz's law is a consequence of **conservation of energy** — the induced current creates a magnetic field that opposes the change causing it, so work must be done to maintain the change. If the induced current aided the change, you would get energy from nothing, violating the first law of thermodynamics.



Methods of inducing an EMF:

- Moving a conductor through a magnetic field
- Changing the area of a circuit in a magnetic field
- Rotating a coil in a magnetic field (this is how generators work)
- Changing the magnetic flux density (e.g., moving a magnet towards a coil)

The AC Generator

For a coil rotating at angular velocity ω in a uniform field B:

$$\Phi = BAN\cos(\omega t) \quad \rightarrow \quad \varepsilon = BAN\omega\sin(\omega t)$$

The maximum (peak) EMF is: $\varepsilon_0 = BAN\omega$

The EMF is zero when the coil is perpendicular to the field (maximum flux, zero rate of change) and maximum when the coil is parallel to the field (zero flux, maximum rate of change).

Worked Example: A coil of 200 turns and area 0.04 m^2 rotates at 50 Hz in a field of 0.30 T. Calculate the peak EMF.

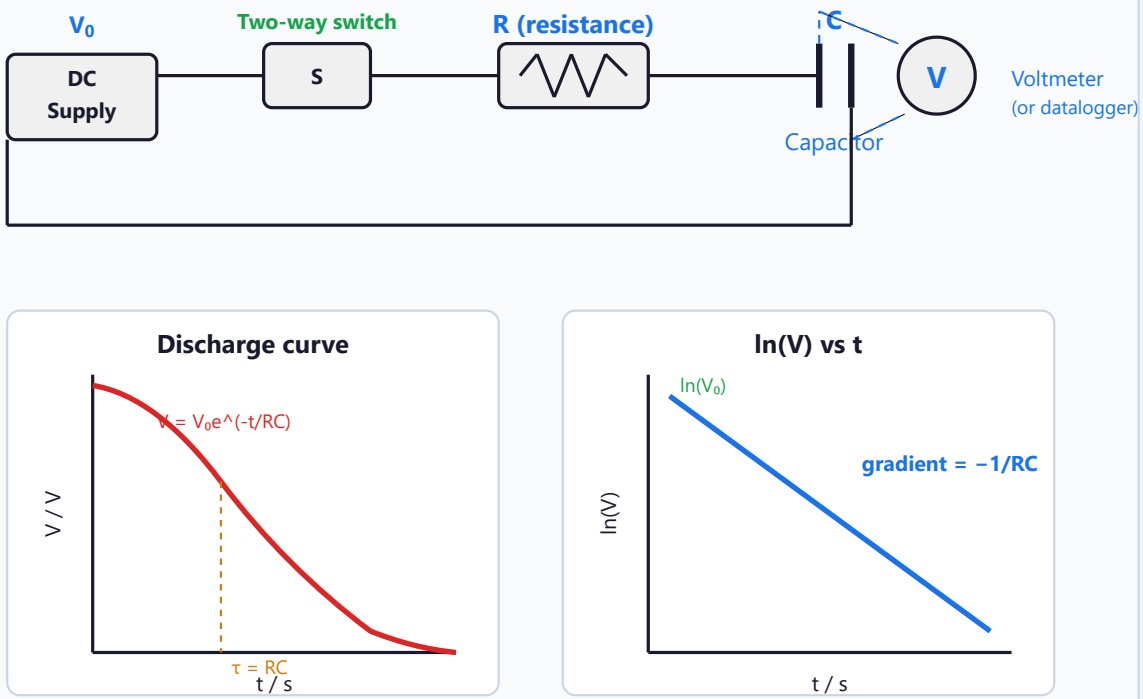
$\omega = 2\pi f = 2\pi(50) = 314$ rad s^{-1}
$\varepsilon_0 = BAN\omega = 0.30 \times 0.04 \times 200 \times 314 =$ **754 V**

Exam Tip: When applying Faraday's law, identify what is changing: is it B, A, or θ? For a rotating coil, $\Phi = BAN\cos(\omega t)$, so $\varepsilon = BAN\omega\sin(\omega t)$. The maximum EMF is $\varepsilon_0 = BAN\omega$.

Summary of Key Quantities

Quantity	Symbol	Unit
Electric field strength	E	N C^{-1} or V m^{-1}
Electric potential	V	V (volts)
Capacitance	C	F (farads)
Time constant	$\tau = RC$	s
Magnetic flux density	B	T (tesla)
Magnetic flux	Φ	Wb (weber)
Flux linkage	NΦ	Wb turns
EMF (induced)	ε	V (volts)

⚗ Core Practical 7 / 15: Investigate capacitor charge and discharge



CP7/15: Capacitor charge and discharge

Apparatus

- Capacitor (e.g. 470 µF or 1000 µF electrolytic)
- Resistor (e.g. 10 kΩ or 100 kΩ — chosen to give RC ≈ 5–50 s)
- DC power supply (e.g. 6 V)
- Two-way switch (or two separate switches)
- Voltmeter or datalogger with voltage sensor
- Stopwatch (if recording manually)
- Connecting wires

Method

1. Connect the circuit: power supply → two-way switch → resistor → capacitor. Connect the voltmeter (or datalogger) in parallel across the capacitor.
2. **Charging:** Flip the switch to connect the power supply. Record the voltage across the capacitor at regular time intervals (e.g. every 5 or 10 s) until it reaches approximately V_0.
3. **Discharging:** Once fully charged, flip the switch to disconnect the supply and allow the capacitor to discharge through R only. Record V at regular time intervals until V is near zero.

4. For the discharge: record at least 10 pairs of (t, V) readings.
5. A datalogger is preferable for accuracy — it records V continuously and can take readings at precise intervals.

Safety

- Use low voltages (≤ 12 V DC).
- Ensure electrolytic capacitors are connected with correct polarity (positive to positive).
- Discharge the capacitor fully before disassembling the circuit.
- Do not use very large capacitors at high voltages — the stored energy could be dangerous.

Analysis

Discharge: The voltage decays exponentially: $V = V_0 e^{-t/RC}$

Taking natural logs: $\ln(V) = \ln(V_0) - t/RC$

Plot **ln(V)** (y-axis) against **t** (x-axis). This gives a straight line with:

- **Gradient** = $-1/RC$
- **y-intercept** = $\ln(V_0)$

From the gradient, calculate RC (the time constant). If R is known, calculate C, and vice versa.

Sources of Error & Improvements

Source of Error	Type	Improvement
Capacitor tolerance (actual C differs from marked value)	Systematic	Measure C independently or determine from the graph
Voltmeter draws current, affecting the discharge rate	Systematic	Use a high-impedance voltmeter or datalogger (input resistance >> R)
Manual timing errors if reading V at intervals	Random	Use a datalogger for automated, precise time-stamped readings

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Capacitor leakage	Systematic	Use a good-quality capacitor; perform experiment quickly
Contact resistance in wires and switch	Systematic	Use short, thick connecting wires; ensure firm connections

Exam Tip: If your ln(V) vs t graph is not a straight line, it suggests the discharge is not purely exponential. This could be caused by the voltmeter having too low an impedance (comparable to R), drawing significant current and affecting the discharge rate.

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